



Loa Loa & Covid19

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Chapter One: Loa loa

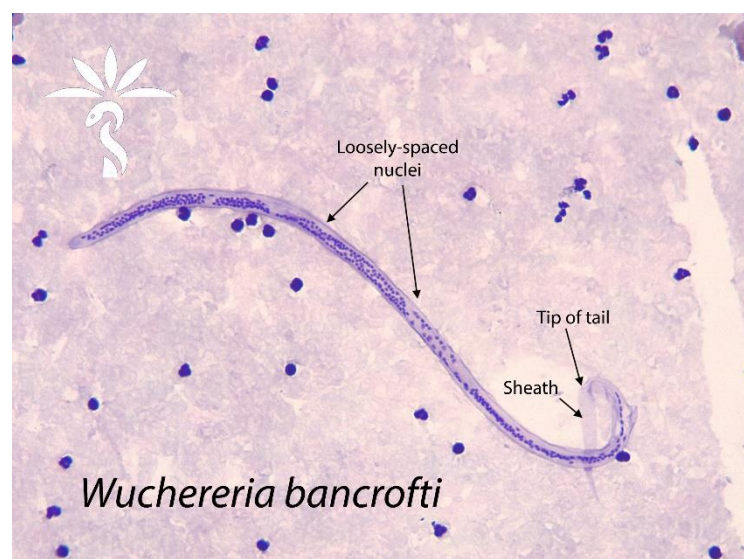
1. Characteristics and Morphology

Loa loa is a tissue-dwelling filarial nematode belonging to the family Onchocercidae. It is the etiological agent of loiasis, a vector-borne helminthic infection endemic to rainforest regions of West and Central Africa.

Morphologically, the adult worm is elongated, cylindrical, and transparent, with a smooth cuticle lacking prominent ridges. Sexual dimorphism is evident: females measure approximately 50–70 mm in length, whereas males are shorter (30–35 mm) and possess a characteristically curved posterior end. The parasite resides primarily in the subcutaneous connective tissues, where it migrates actively.

The microfilarial stage is sheathed and measures approximately 250–300 μm in length. A key diagnostic feature is the continuous extension of nuclei to the tip of the tail, which differentiates it from other sheathed filarial species.

Notably, *Loa loa* microfilariae exhibit diurnal periodicity, with peak peripheral blood concentration occurring between 10:00 AM and 2:00 PM. This periodicity reflects an evolutionary adaptation synchronized with the daytime feeding behavior of its vector, the deer fly of the genus *Chrysops*.



2. Source of Infection

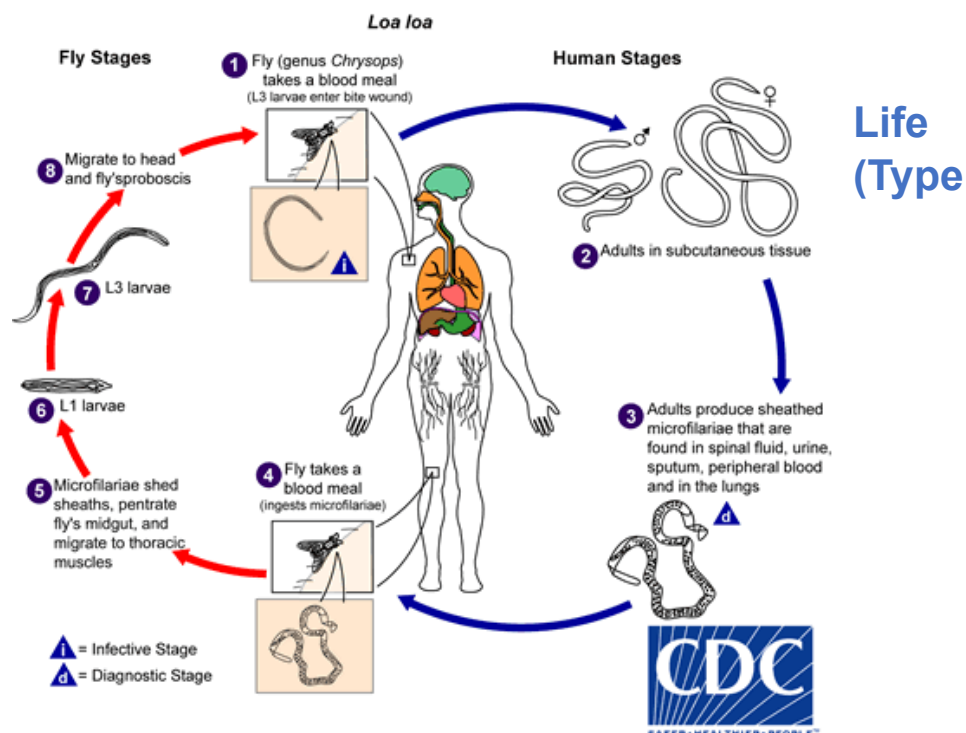
The primary source of infection is an infected human host harboring circulating microfilariae in the peripheral blood. Humans serve as the principal and epidemiologically significant reservoir. Unlike some other filarial parasites, animal reservoirs do not play a major role in maintaining transmission cycles.

3. Mode of Transmission

Transmission occurs through the bite of infected *Chrysops* flies. During a blood meal, the vector ingests circulating microfilariae. Within the thoracic muscles of the fly, the microfilariae undergo developmental transformation from first-stage larvae (L1) to infective third-stage larvae (L3) over approximately 10–12 days. Importantly, this represents a cyclodevelopmental transmission pattern, in which the parasite develops but does not multiply within the vector.

When the infected fly takes another blood meal, L3 larvae migrate to the proboscis and are deposited onto the human skin, entering through the bite wound. Once inside the human host, the larvae migrate through subcutaneous tissues and gradually mature into adult worms over several months.

4. Cycle



of Life Cycle)

Loa loa has an indirect (heteroxenous) life cycle requiring two hosts: a definitive human host and an intermediate insect vector. Sexual reproduction occurs exclusively in the human host, while larval development to the infective stage occurs within the vector. The parasite may survive for many years in the human body, and adult worms can live for over a decade, sustaining chronic infection. Schematic overview of neurotransmitter-related pathways implicated in major neurological disorders.

5. Hosts (Type of Hosts)

The definitive host is the human, in whom adult worms reach sexual maturity and produce microfilariae. The intermediate host and biological vector is the *Chrysops* fly, which facilitates development to the infective stage and transmission to new hosts. events and investigation of NT dysregulation, linking detection capabilities directly to understanding disease mechanisms and developing therapeutic strategies.

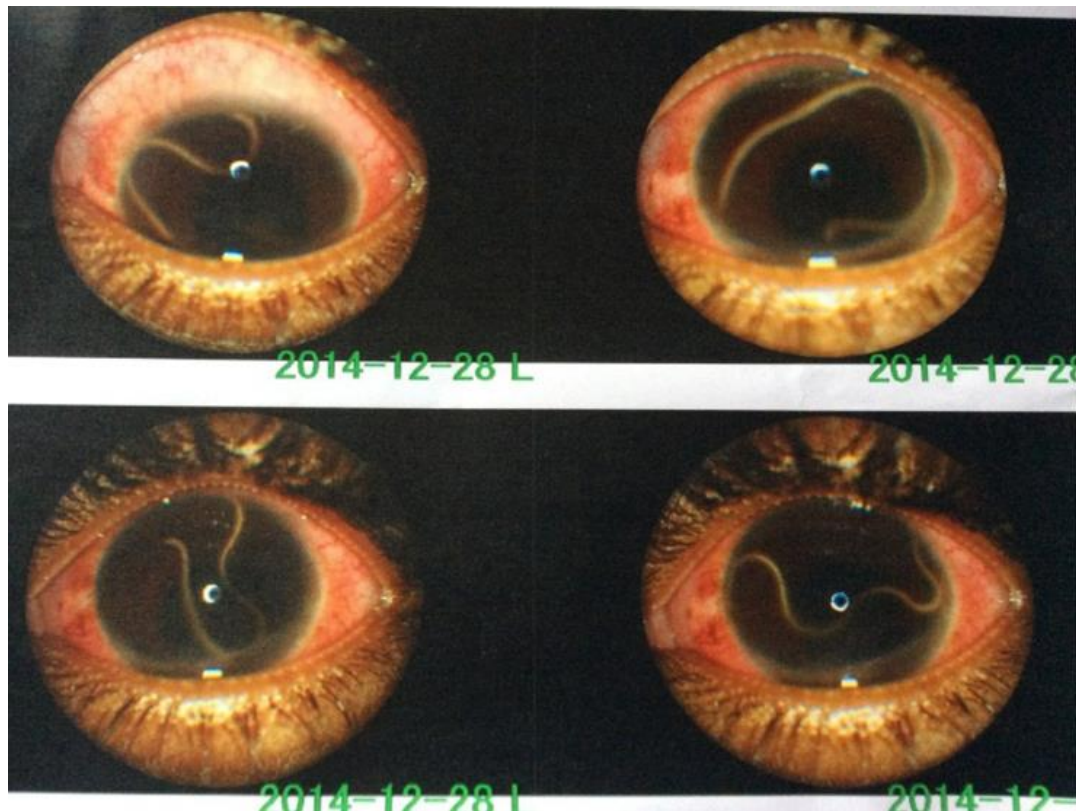
6. Pathogenesis

The pathogenesis of loiasis is predominantly immune-mediated rather than directly destructive. Adult worm migration through subcutaneous tissues induces localized inflammatory reactions characterized by mast cell activation, histamine release, and eosinophil recruitment. These immune responses culminate in the formation of transient angioedematous swellings known as Calabar swellings.

Immunologically, infection induces a Th2-dominant response, characterized by elevated interleukins IL-4, IL-5, and IL-13. These cytokines promote IgE production and eosinophil proliferation. Eosinophils contribute to host defense through antibody-dependent cellular cytotoxicity but also mediate tissue inflammation. Chronic infection may result in partial immune tolerance, especially in endemic populations with high microfilarial loads.

A particularly serious complication occurs following treatment in patients with high microfilaremia. Rapid parasite death can trigger massive antigen release, complement activation, disruption of the blood–brain barrier, and inflammatory encephalopathy. This represents an immunopathological phenomenon rather than direct central nervous system invasion.

Laboratory diagnosis relies on daytime peripheral blood smear examination using Giemsa stain, taking into account the diurnal periodicity of microfilaremia. Molecular methods such as PCR



Conclusion

Loiasis exemplifies a sophisticated host–parasite interaction involving circadian adaptation, immune modulation, and hypersensitivity phenomena. The clinical manifestations reflect the balance between parasite persistence and host immune reactivity.

Chapter Two: COVID-19

1. Classification

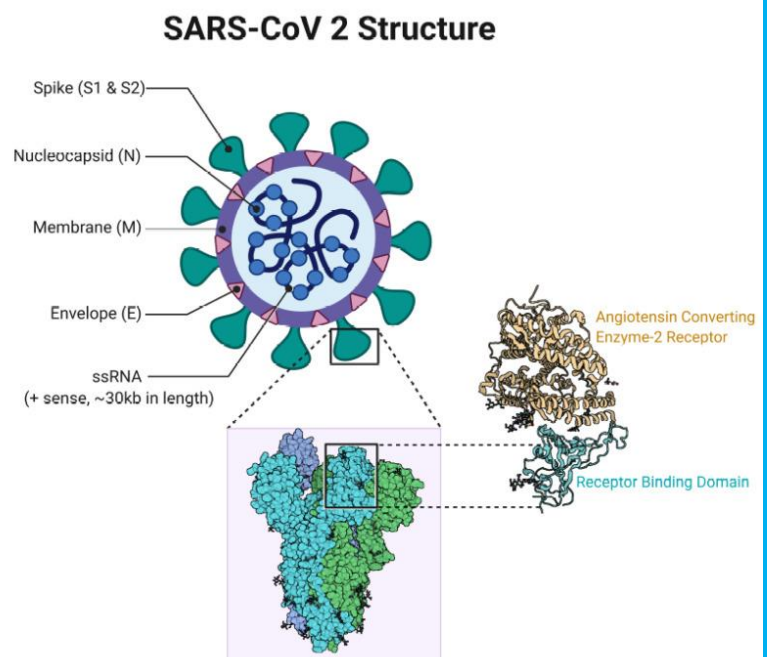
COVID-19 is caused by SARS-CoV-2, a member of the genus Betacoronavirus within the family Coronaviridae. It belongs to the subgenus Sarbecovirus and is genetically related to SARS-CoV-1. The virus possesses a positive-sense single-stranded RNA genome and is enveloped.

2. Characteristics and Morphology

SARS-CoV-2 is spherical to pleomorphic, measuring approximately 80–120 nm in diameter. Its defining feature is the presence of surface spike (S) glycoproteins that give the virion a crown-like appearance under electron microscopy.

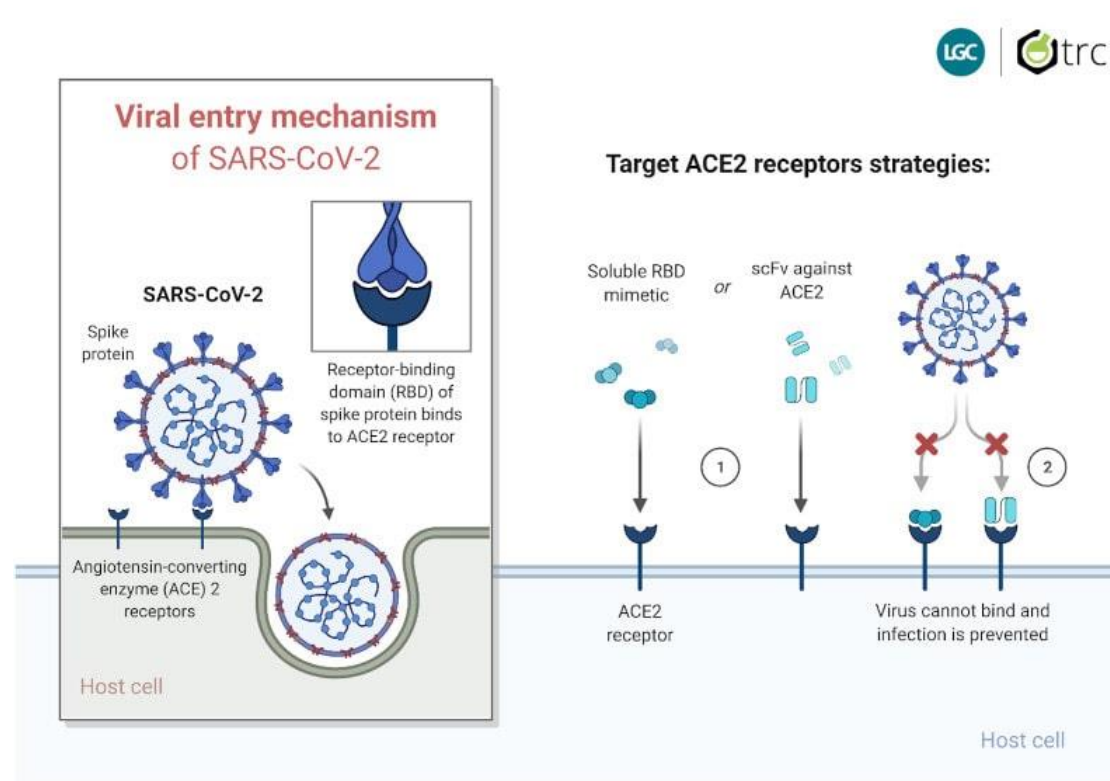
The viral genome is approximately 30 kilobases in length, making it one of the largest RNA virus genomes. It contains open reading frames (ORF1a and ORF1b) encoding non-structural proteins that form the replication-transcription complex. Structural proteins include the spike (S), membrane (M), envelope (E), and nucleocapsid (N) proteins.

The S protein mediates viral entry by binding to the angiotensin-converting enzyme 2 (ACE2) receptor. The S1 subunit contains the receptor-binding domain, whereas the S2 subunit facilitates membrane fusion following proteolytic cleavage by host proteases such as TMPRSS2.



3. Mode of Transmission

Transmission occurs primarily through respiratory droplets and aerosols generated during coughing, speaking, or breathing. The virus gains entry through inhalation and binds to ACE2 receptors expressed abundantly on nasal epithelium and type II pneumocytes. The distribution of ACE2 in multiple tissues, including vascular endothelium and gastrointestinal epithelium, explains the systemic manifestations of the disease.



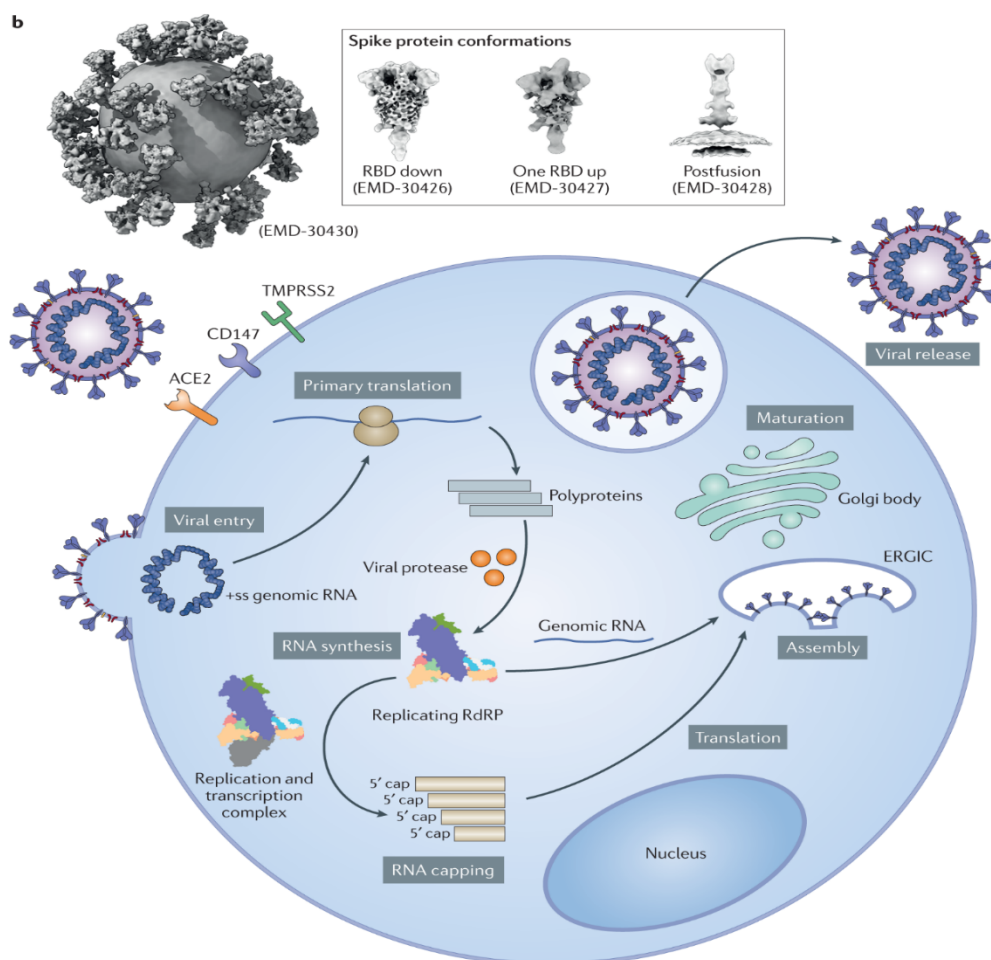
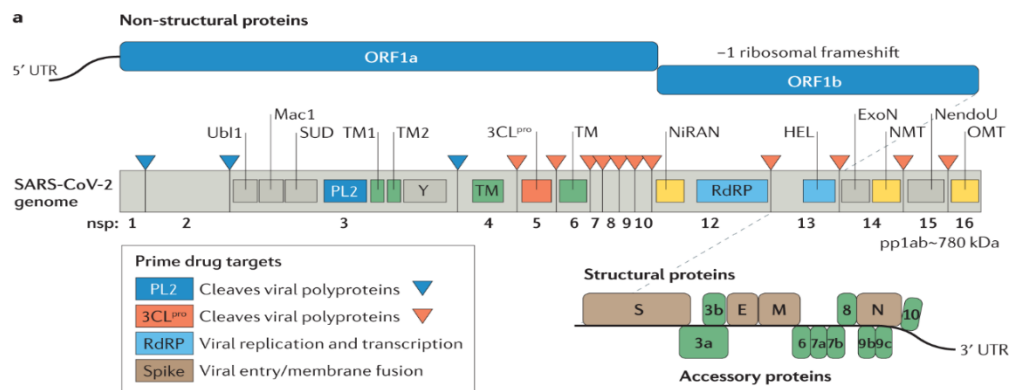
4. Replication

Replication occurs entirely within the cytoplasm. After attachment and entry, the viral RNA functions directly as messenger RNA. Translation of ORF1a and ORF1b produces polyproteins that are cleaved into functional non-structural proteins, including the RNA-dependent RNA polymerase (NSP12). These proteins assemble into a replication-transcription complex within double-membrane vesicles derived from the endoplasmic reticulum.

The polymerase synthesizes a negative-sense RNA template, which is used to generate genomic and subgenomic RNAs.

Structural proteins are translated and inserted into the endoplasmic reticulum and Golgi apparatus. Virion assembly occurs in the ER-Golgi intermediate compartment, followed by release via exocytosis.

Notably, SARS-CoV-2 possesses a proofreading mechanism mediated by NSP14, which enhances replication fidelity and contributes to its relatively large genome size compared to other RNA viruses.

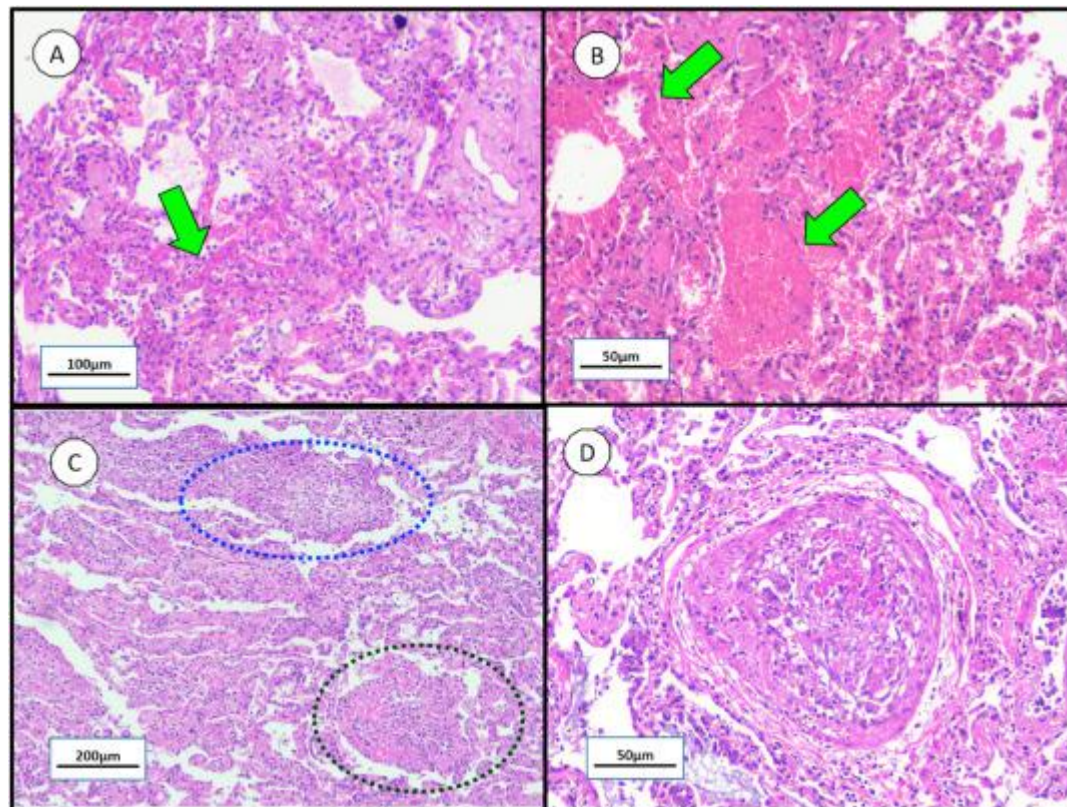


5. Clinical Manifestations

Clinical presentation ranges from asymptomatic infection to severe respiratory failure. Early symptoms commonly include fever, cough, and anosmia. Anosmia results primarily from infection and dysfunction of sustentacular cells in the olfactory epithelium rather than direct neuronal destruction.

Severe cases are characterized by viral pneumonia and acute respiratory distress syndrome (ARDS). Excessive immune activation leads to a cytokine storm involving IL-6, TNF- α , and other pro-inflammatory mediators. This hyperinflammatory state increases vascular permeability, causing diffuse alveolar damage and impaired gas exchange.

Endothelial injury and activation of coagulation pathways result in microthrombi formation, explaining elevated D-dimer levels and thromboembolic complications.



6. Diagnosis

The gold standard for diagnosis is reverse transcription polymerase chain reaction (RT-PCR), which detects viral RNA targeting genes such as N, E, or RdRp. Cycle threshold (Ct) values provide indirect estimation of viral load. Antigen detection tests offer rapid results but have lower sensitivity. Imaging studies may reveal bilateral ground-glass opacities consistent with viral pneumonia.

Conclusion

COVID-19 represents a complex viral disease in which both direct cytopathic effects and dysregulated host immune responses contribute to pathology. Its replication strategy, immune evasion mechanisms, and thromboinflammatory complications underscore the intricate interplay between viral biology and host defense.

References

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2. Bennett JE, Dolin R, Blaser MJ. Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases. Elsevier.